

Passive and active cavitation detection methods: inception and persistence during HIFU sonication

Gonzalo Garay

Laboratorio de Acústica Ultrasonora
Facultad de Ciencias
Montevideo, Uruguay
ggaray.uy@gmail.com

Yamil Abraham

Laboratorio de Acústica Ultrasonora
Facultad de Ciencias
Montevideo, Uruguay

Guillermo Cortela

Laboratorio de Acústica Ultrasonora
Facultad de Ciencias
Montevideo, Uruguay

Nicolás Benech

Laboratorio de Acústica Ultrasonora
Facultad de Ciencias
Montevideo, Uruguay

Carlos Negreira

Laboratorio de Acústica Ultrasonora
Facultad de Ciencias
Montevideo, Uruguay

Abstract—High-intensity focused ultrasound (HIFU) is important for its application in clinical therapies. However, the phenomenon of acoustic cavitation associated with HIFU sonication in biological tissues must be carefully controlled. In this work, we aimed to investigate two aspects of this phenomenon: the generation of bubble clouds by acoustic cavitation and their evolution over time. For this study, we applied HIFU to an agar-agar phantom with exposure time of 0.6 s and driving frequency of 2 MHz. To monitor cavitation activity, we utilized both a passive and active method. A passive cavitation detector (PCD) was placed perpendicular to the HIFU transducer beam, allowing us to measure the wave from the cavitation cloud during HIFU application. Additionally, we utilized a 64-channel array to scan the phantom once HIFU ceased. Results show that both methods complement each other for the monitoring of the cavitation activity at different times during HIFU sequence. This provides information about the inception and persistence of the phenomenon, which can aid in the control and optimization of HIFU applications.

Index Terms—Cavitation detection, HIFU.

I. INTRODUCTION

The use of high-intensity focused ultrasound (HIFU) has gained importance for its application in clinical therapies. HIFU is capable of delivering a precise amount of energy to rapidly increase the temperature in a tissue while minimizing its effect on the surrounding vasculature. For this, HIFU has become a safe method to ablate tumors which are close to major blood vessels [1]. Applications in histotripsy require the mechanical disintegration of soft tissues. It has been demonstrated that this is due to HIFU-induced bubble activity [2]. On the other hand, studies aiming to investigate the opening of the blood-brain barrier during focused ultrasound application require the monitoring of stable cavitation activity of microbubbles. However, the origin of transient cavitation is undesirable [3]. In either case, the phenomenon of acoustic cavitation associated with HIFU sonication in biological tissues must be carefully controlled.

In this work, our objective is to monitor both the inception and persistence of cavitation activity using two distinct

methodologies. The first involves a passive detection approach utilizing a passive cavitation detector (PCD), while the second employs an active detection strategy relying on the acquisition of ultrasound images through a pulse-echo technique. This study is conducted during HIFU sonication of an agar-agar based phantom.

For the purposes of this study, we aim to investigate the cavitation activity over an extended exposure time (0.6 s). The primary objective is to examine the complementarity of the two methods of detection and the information that can be derived from their combined application. The passive method allows us to study cavitation during its inception, while the active method enables us to investigate the activity once HIFU ceases.

II. METHOD

The experimental setup is shown on figure 1. A dual configuration consisting on a HIFU probe and a linear array (HIFUPlex-03, Sonic Concepts) sonicates an agar-agar phantom. The PCD (Precision Acoustics) senses the wave coming from the phantom. This setup is submerged in degassed water.

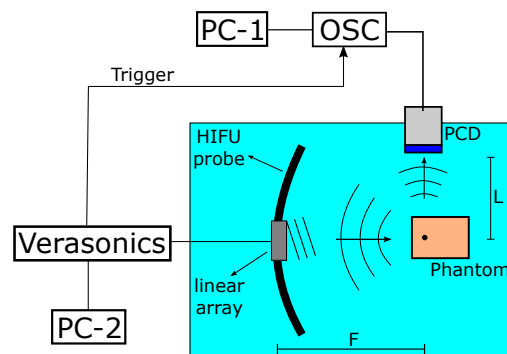


Fig. 1. Experimental setup. A HIFU probe and a linear array sonicates an agar-agar phantom. The PCD senses the wave coming from the phantom. This setup is submerged in degassed water.

The HIFU probe is a spherical array composed of 8 concentric rings of nominal frequency 2 MHz. The outer diameter of the full probe is 64 mm and the inner diameter is 32 mm. In the experiment the focal distance F is the geometrical focus of the probe which is $F = 63.2$ mm. The linear array is positioned at the center of the HIFU probe to scan the sonicated area and acquire a series of ultrasound images. It is a set of 64 transducers of nominal frequency 5 MHz. The total length is 10 mm and the pitch is 0.15 mm.

The phantom is a cylindrical sample (diameter 60 mm, height 70 mm) of a solidified solution (1.75% w/v) of agar-agar powder and water [4]. It was characterized by measuring its density ρ , longitudinal wavespeed c , attenuation coefficient α at 1 MHz, and the nonlinear parameter B/A . These parameters were measured relatively to water parameters assumed from refs. [5], [6], [7] and [8]. Density was measured through a hydrostatic weighing. The longitudinal wavespeed and the attenuation coefficient was measured through ultrasonic transmission by measuring the time of flight and the amplitude attenuation [9]. The B/A parameter was measured by a finite amplitude insert-substitution method (FAIS) [8]. The parameters are summarized in Table I. In the experiment, the phantom is positioned in such a way that the wave emitted by the HIFU probe impacts one of its flat faces, and the focal point is located 1 cm inside.

The PCD is a flat transducer of diameter 23 mm and nominal frequency 2 MHz. It is positioned in a con-focal arrangement at a distance $L = 88$ mm from the focal point. The signal acquired is digitalized in the oscilloscope (OSC) and stored in the first computer (PC-1).

In the experiment, the HIFU sonication and the linear array scanning are controlled with a programmable Verasonics Vantage system. The HIFU probe is excited with a sine signal of frequency $f_0 = 2$ MHz, duration 0.6 s and amplitudes $V_{in} = 14$ V and $V_{in} = 18$ V. At this levels, the estimated pressure in the focal position in water are 12 MPa and 16 MPa, respectively. The linear array elements are excited with 2 cycles of a sine function of central frequency 5 MHz. The acquired ultrasound images are based on a coherent plane-wave compounding method [10]. For this, delays are introduced in the emission of each transducer so that 24 plane waves are emitted at intervals of $500 \mu\text{s}$, sweeping an angle of 60° centered on the array axis. After that, the signals acquired are beamformed and stored on the second computer (PC-2).

The ultrasonic sequence can be described as follows. First, a background image is acquired to sense the state of the phantom before HIFU sonication. After 12.5 ms the HIFU sonication starts and the acquisition on the PCD is triggered to acquire the

arrival of the wave that comes from the phantom. 0.50 s after HIFU sonication ends, 22 images are acquired at intervals of 0.51 s. The time T of each frame is defined at the beginning of the image acquisition process. 20.5 s after the background acquisition, the same procedure is repeated one more time. This sequence is performed with the amplitude $V_{in} = 14$ V. Then, the phantom is laterally displaced more than 2 mm, a distance higher than the focal width, and the sequence is repeated with $V_{in} = 18$ V.

III. RESULTS

First of all, we analyze the results obtained with the amplitude $V_{in} = 14$ V (12 MPa in water) on the HIFU probe. The results of the active cavitation are shown in Fig. 2. The background image ($T = 0$) and frames at $T = 1.1$ s and $T = 11.9$ s are shown for both the first and the second HIFU sonication. Comparing Figs. 2(a) and 2(b), we can observe a cavitation spot forming around the focal point after sonication. This spot almost completely disappears after some time, so that it is no longer visible in Fig. 2(c). Despite what was stated, a small spot at coordinates $(x, z) = (0, 50)$ mm remains. The second background image [Fig. 2(d)], prior to repeating the sonication, shows that this tiny spot persists. Following the second sonication [Fig. 2(e)], there is no evidence of new cavitation events. In the last displayed frame [Fig. 2(f)], the phantom has fully recovered.

The results of passive cavitation detection are shown in Fig. 3. The signal amplitudes are normalized with respect to the maximum amplitude obtained in the entire experiment, which is achieved with a higher voltage. During the first sonication, the wave is scattered from the bubbles formed around the focal point reaching the PCD. The corresponding signal is shown

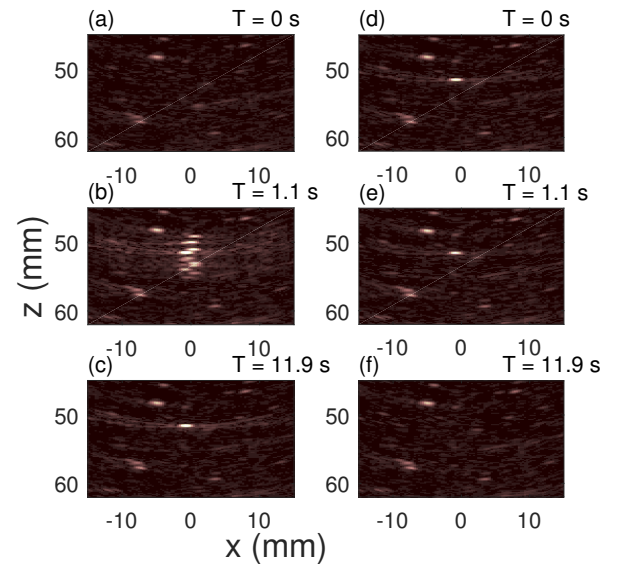


Fig. 2. Ultrasound images of the phantom around the focal point for a voltage amplitude $V_{in} = 14$ V. First sonication at (a) $T = 0$ s, (b) $T = 1.1$ s and $T = 11.19$ s. Second sonication at (a) $T = 0$ s, (b) $T = 1.1$ s and $T = 11.19$ s. A spot is observed at $T = 1.1$ s during first sonication.

TABLE I
WAVE PROPAGATION PROPERTIES AT 20°C AND 1 MHz.

	ρ (g/cm ³)	c (mm/ μs)	α (dB/cm)	B/A
water ^a	0.998	1.485	0.0022	5.2
phantom	1.001	1.489	0.0292	7.2

^aFrom refs. [5], [6], [7] and [8].

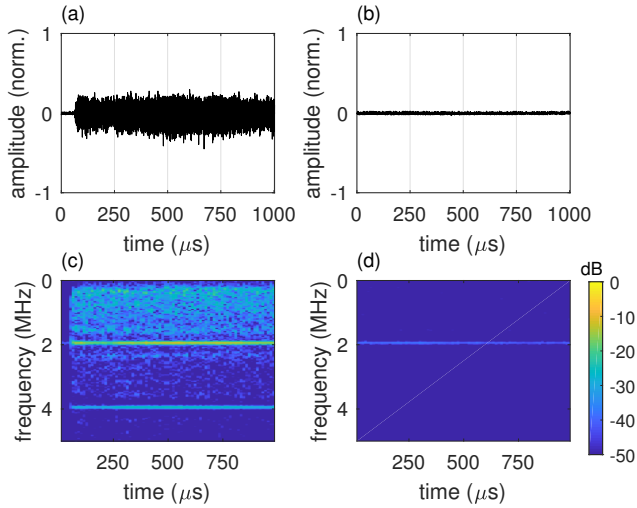


Fig. 3. Signals acquired in the PCD with $V_{in} = 14$ V during (a) the first sonication and (b) the second sonication. (c)(d) Its respective spectrograms. A wave is observed during first sonication. During second sonication noise is registered.

in Fig. 3(a). During the second sonication, due to the absence of an equivalent cavitation event and no scattering occurring, the PCD only records noise [Fig. 3(b)].

The signal spectra are analyzed through a spectrogram, which involves splitting the signal into windows of $25 \mu\text{s}$ (50 cycles) with a 50% overlap and performing a fast Fourier transform (FFT) on each window. Spectrograms are shown in a color-scale that represent the intensity level in dB. The 0 dB level is at the maximum intensity obtained in the entire experiment. Fig. 3(c) displays the spectrogram corresponding to the signal in Fig. 3(a). It shows the fundamental harmonic at a frequency of 2 MHz, the second harmonic at 4 MHz, and a "background noise" that reaches a higher level at frequencies lower than the fundamental frequency. The spectrogram of the measured signal after the second sonication [Fig. 3(d)] only shows the fundamental frequency with a very low intensity level.

Secondly, we analyzed the results obtained with $V_{in} = 18$ V (16 MPa in water). The result of active cavitation detection is shown in Fig. 4. In this case, a larger-sized spot is observed immediately following the first sonication [Fig. 4(b)]. This spot persists notably until the final frame [Fig. 4(c)]. It remains visible even in the second background image [Fig. 4(d)] and continues to be present after the second sonication [Figs. 4(e) and 4(f)].

The waves measured in the PCD are depicted in Fig. 5. During the first sonication [Fig. 5(a)], a signal of high amplitude is obtained, the maximum achieved throughout the experiment. Its spectrogram [Fig. 5(c)] shows characteristics similar to those obtained with a lower voltage [Fig. 3(c)]. After the second sonication, the signal shows a stable amplitude [Fig. 5(b)], and its spectrogram shows some well-defined frequencies below the fundamental frequency across

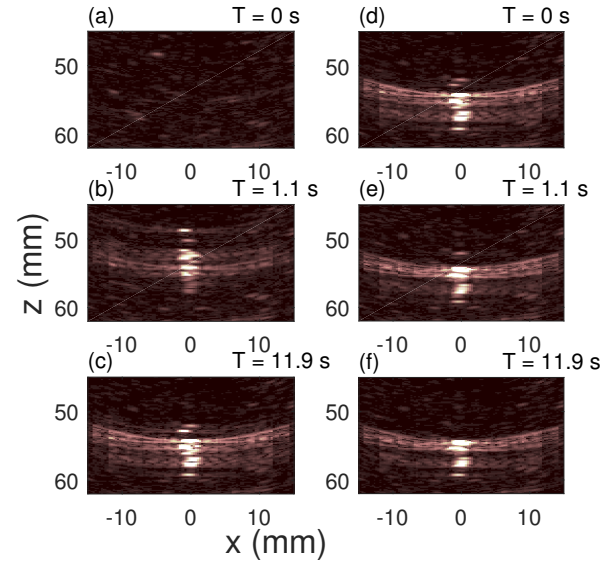


Fig. 4. Ultrasound images of the phantom around the focal point for a voltage amplitude $V_{in} = 18$ V. First sonication at (a) $T = 0$ s, (b) $T = 1.1$ s and $T = 11.19$ s. Second sonication at (a) $T = 0$ s, (b) $T = 1.1$ s and $T = 11.19$ s. A spot is observed at $T = 1.1$ s and persists until the last frame shown.

the displayed temporal range [Fig. 5(c)].

In Fig. 6, the spectra at $750 \mu\text{s}$ are plotted, obtained from the spectrograms shown in Figs. 5(c) and 5(d). In both cases, the intensity level at the fundamental frequency is similar (difference of approximately 2 dB). However, at frequencies below the fundamental one, during the first sonication (blue line), the signal reaches higher intensity than during the second one (red line). Furthermore, in this latter case, well-defined peaks are observed at frequencies that are multiples of $f_0/6$.

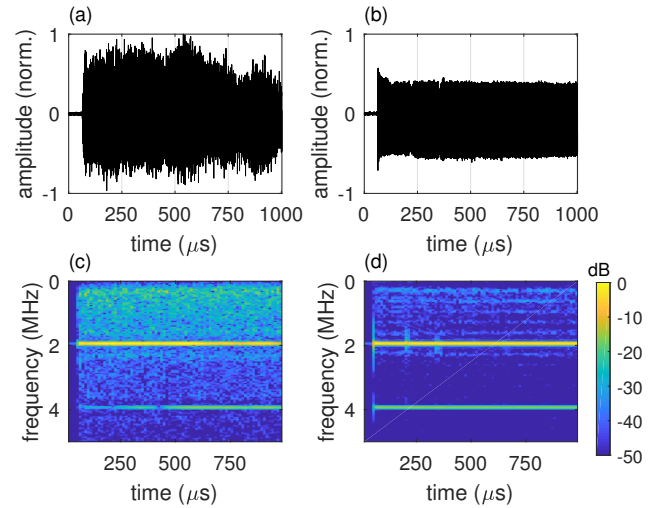


Fig. 5. Signals acquired in the PCD with $V_{in} = 18$ V during (a) the first sonication and (b) the second sonication. (c)(d) Its respective spectrograms. A wave is observed during both sonications. The second one shows a stabilized amplitude.

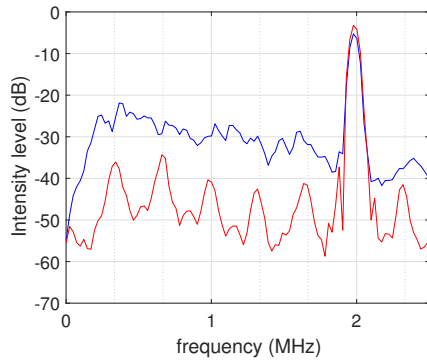


Fig. 6. Spectra of the signals measured during with $V_{in} = 18$ V during the first sonication (blue line) and the second sonication (red line). The first case shows a continuous spectrum in frequencies below 2 MHz. The second case shows peaks at frequencies multiples of $f_0/6$.

IV. DISCUSSION

The results show that the two employed voltages amplitudes are sufficient to generate cavitation. Bubble nucleation is primarily initiated by tensile pressures [11], and in both cases, the attained pressure is sufficient to reach the required threshold for the event to occur. The difference lies in the duration of the bubble cloud after the HIFU sonication is finished. We believe this is due to heating. This effect forms the basis of boiling histotripsy and occurs with extended HIFU exposures [2]. Because the exposure times in this experiment are quite lengthy, a temperature higher than ambient is sustained for a longer period, facilitating the persistence of the bubbles.

Comparing the signals obtained in the PCD with both voltages [Figs. 3(a) and 5(a)], we observe that the amplitude increases by approximately five times when the voltage in the HIFU probe is raised. This increase does not correspond directly to the applied voltage increment. Therefore, we believe it is due to an enlargement in the bubble cloud. This aligns with the increase in the spot size observed in the images.

Let's now compare what occurs between the first sonication and the second one. With $V_{in} = 14$ V, we find that no cavitation occurs during the second sonication. This is consistent with reported results that show a reduction in the probability of generating cavitation from one ultrasound exposure to another [12]. In the experiments presented here, we believe that the probability of generating cavitation is greatly reduced due to the long exposure times employed.

At $V_{in} = 18$ V, cavitation activity is recorded during the second sonication. In this case, the recorded signal has a significantly more stable amplitude than during the first sonication. The spectrum shows peaks at frequency multiples of $f_0/6$. The appearance of subharmonics is associated with the nonlinear oscillation of the bubbles [13] and is linked to stable cavitation. Especially, the $f_0/2$ subharmonic is considered as an acoustic signature of the bubble [3]. In this work, we think that the appearance of these subharmonics is due to the stable oscillation of the bubbles, as the second sonication is performed on bubbles previously formed.

V. CONCLUSION

In this work, the active cavitation detection method and the passive method operate independently to observe bubble activity. Nevertheless, these methods work together to complement each other. The passive method enables the examination of signal characteristics originating from the cavitation bubble during HIFU application. This provides insights into spectral components. On the other hand, the active method facilitates the localization of the region where the bubble cloud forms and provides at least qualitative information about its size.

The observations made about the size and persistence of the bubble cloud allow us to gather some information about the thermal effects around the focal zone. This implies that, for extended HIFU exposures, cavitation and heating effects must be studied together. Therefore, we intend to utilize the experimental setup presented here to collect information about the thermal behavior of different phantoms based on cavitation activity.

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